

Akhil Chigurupati

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PROFESSIONAL SUMMARY

Software engineer who builds and ships production software, then owns what happens after it goes live. Experience spans **FastAPI** backend services, **React** frontends, webhook-driven payment integrations, and **AI agent pipelines**. Designed, prompted, and iterated on agents that handle real tasks end-to-end with minimal human intervention. I believe in making it exist first and then improving it. Heavy user of **Claude**, **Groq**, and **Cursor**, building faster without cutting corners on what ships. Comfortable reading third-party API docs, wiring webhooks, and demoing working software every week.

PROFESSIONAL EXPERIENCE

Tata Consultancy Services | Software Engineer | Apr 2023 – May 2024

- Built Python **microservices** and **REST APIs** for a SaaS monitoring platform managing **500+ production services**, reducing debug cycle time through clean service boundaries and cutting manual incident resolution by **30-40%**.
- Architected **Apache Kafka** event-driven pipelines processing **50,000+ events/min** for real-time log ingestion, cutting alert latency from minutes to seconds and enabling pre-impact incident detection across distributed services.
- Integrated **AI/ML anomaly detection models** into production workflows across 500+ services, flagging failure patterns **8-12 minutes** before user-facing degradation; owned model integration, production rollout, and post-deployment monitoring.
- Deployed on **Docker** and **Kubernetes** across **1,000+ pods** with auto-scaling and **Prometheus/Grafana** observability; automated remediation workflows and stayed on-call owning root-cause resolution end-to-end.

PROJECT EXPERIENCE

Sales Signal Intelligence Platform | sales-signal-platform.vercel.app/

FastAPI • React • TypeScript • PostgreSQL (Neon) • HubSpot API • Groq • Claude API • GitHub Actions

- Classified 100% of 50 HubSpot deals** across **7 pipeline stages** (At Risk / Stalling / On Track) using **Claude API** signal analysis of deal age, stage velocity, and contact activity, eliminating manual sales review entirely.
- Engineered a **time-series deal snapshot system** in **PostgreSQL** capturing stage velocity history across 50 deals, enabling **AI pattern detection** unavailable in native HubSpot.
- Optimized sequential **HubSpot API** lookup and **AI classification pipeline** to **~2s per deal (98.9s for 50 deals)**, triggered via a single API call with automated full-pipeline sync.
- Shipped a **React/TypeScript** Kanban dashboard with 7 stage columns, search/filter, and click-to-expand deal modals surfacing **AI-generated risk signals** in real time.

OpenClaw | Multi-Agent AI System | jarvis-plum-gamma-14.vercel.app/

Python • Anthropic MCP SDK • Claude API • LLaMA 3 • Groq API • Whisper • ElevenLabs • FastAPI • PostgreSQL • React • TypeScript • Docker

- Orchestrated 5 parallel specialized agents** (Research, Email, Writer, Code, Monitor) via a **Claude**-powered orchestrator that decomposes voice commands into subtasks, achieving end-to-end latency under **2 seconds**.
- Wired agents to a custom **MCP tool server** exposing browser control, email, **GitHub**, and file operations as callable **AI tools**; persisted every agent decision to **PostgreSQL** for full auditability and replay.
- Built a live **React** dashboard showing agent activity and task decisions in real time; implemented voice-in/voice-out using **Whisper** and **ElevenLabs** with sub-2-second command-to-action response.

AI Incident Intelligence Platform | Event-Driven SaaS

Python • FastAPI • LangChain • LlamaIndex • AWS (SQS, Lambda, DynamoDB, SNS) • React • TypeScript • pgvector • RAGAS • MLflow • Docker • GitHub Actions

- Ingested real-time signals via **AWS SQS** through a **LangChain + LlamaIndex** RAG analysis layer and synced structured outputs downstream - idempotent, queue-buffered, and fully observable.
- Implemented a **RAGAS + MLflow** evaluation pipeline automatically scoring every **LLM** output for faithfulness and relevancy, gating releases on measurable quality thresholds.
- Built server-side event tracking and a **React/TypeScript** frontend with live **SSE** streaming; owned funnel instrumentation from event definition through to iterative improvement.

API to MCP Converter | Agentic Developer Tooling

Python • Anthropic MCP SDK • FastAPI • OpenAPI/Swagger • Docker • GitHub Actions

- Built a CLI tool that reads any **OpenAPI/Swagger** spec and auto-generates a production-ready **MCP server**, exposing every REST endpoint as a callable **AI tool** - reducing REST-to-agent integration from hours to a single command.
- Validated across **10+ public APIs** including Stripe and GitHub; packaged with **Docker** and **GitHub Actions** CI for immediate install and production-ready deployment.

EDUCATION

Regis University - Master of Science, Information Systems | Denver, CO | Aug 2024 – May 2026

Jawaharlal Nehru Technological University - Bachelor of Technology, Computer Science & Information Technology | Hyderabad, India | 2020 – 2024

TECHNICAL SKILLS

Languages & Frontend: Python, TypeScript, JavaScript, SQL, HTML, CSS | React.js, Next.js, Vite, SSE

Backend & Integrations: FastAPI, REST APIs, Webhooks, OpenAPI/Swagger, Microservices, Apache Kafka, AWS SQS, idempotent event processing

AI & Agents: LangChain, LlamaIndex, MCP SDK, LLM Tool-Use, Function Calling, RAG, RAGAS, MLflow, Claude API, OpenAI API, Prompt Engineering

AI Dev Tools: Claude Code, Claude Cowork, Claude Design, Cursor, GitHub Copilot, Groq

CRM & ERP: HubSpot API, CRM deal pipeline integration, deal stage classification, SuiteQL

Cloud & DevOps: AWS (SQS, Lambda, DynamoDB, SNS, S3), Docker, Kubernetes, GitHub Actions, CI/CD, Linux

Databases: PostgreSQL, Redis, Supabase pgvector, MongoDB